Please consider the following use cases.

Use case 1: Chat analytics for early intervention as per Biopsychosocial Model

Aim: To identify mental health factors corresponding to biology, psychology and social factors from user conversations. The model to predict intervention is a learnt pattern from reddit mental health posts.

Expected Outcome: Prognostic model to detect user trajectory and recommend early intervention

Expected Clinical Benefits:

1. Tailoring of Treatment for preemptive measures / early interventions
2. Clarity of Diagnosis - root cause and factors associated to the mental state
3. Object Baseline - the starting point of the journey
4. Risk assessment - how bad the situation is right now

Execution Plan:

1. Retrospective Analysis - Develop user journeys from the data, with subreddit transition times, and subreddits as states.
2. Thematic analysis - Identify the themes / factors associated to each state in the user journeys
3. Prognostic predictive modelling

Example Use case Implementation:

Introduction:

The idea is given a new user, predict the journey of the user and if the user is accelerating faster in states then the clinical intervention is produced given the probability of the path. The first subreddit is considered as the behaviour baseline. The subreddit journey is a graph with hours between each subreddit (defined as “a state” here) and weights (defined as transition time here). The transition between subreddits could be considered as acceleration. Slower the acceleration the better.

Dataset used in the use case is Reddit Mental health dataset
https://www.kaggle.com/datasets/entenam/reddit-mental-health-dataset

Data Preprocess:

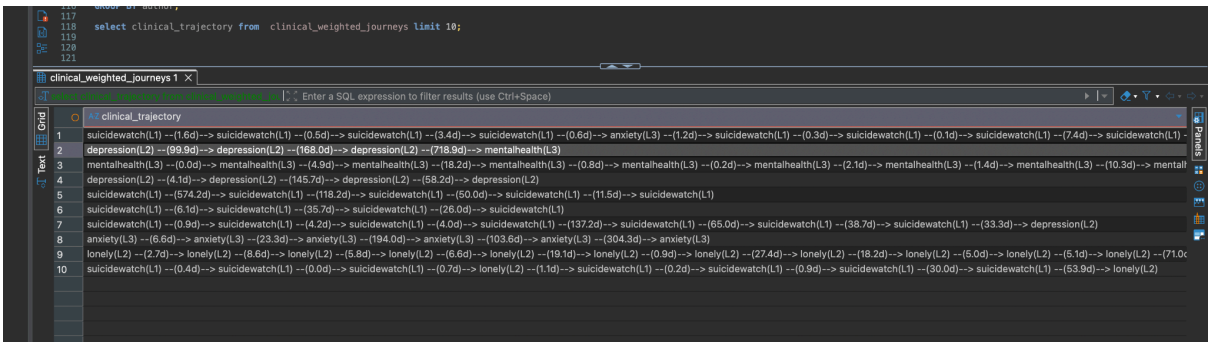
This step chronologically aggregates user information to form longitudinal clinical journeys.
A journey here is referenced as the shift of mental state (if any) over the period of time of a user. The total number of records is 1, 797, 016. Distinct users are 838, 282.

Following are the steps performed for the ETL.

1. Loaded the csv files from the folders and loaded them into the staging table in postgreSQL.
2. Created a user journey base table by partitioning the table as per the author, generating the event index where the event is the post published. Calculated the lag value of previous post time, and previous subreddit.

3. Extracted temporal features like day of week, month and hour from the timestamp column.
4. Analysis is performed to develop the user journey (serialised transition model). Unique subreddits are identified and classified as levels based on escalation scale referred from Hierarchical Risk Model (HiTOP) defined in the field of psychiatry.

Level	Severity	Subreddits	Clinical Interpretation
Level 3	**Crisis**	suicidewatch	**Acute Risk:** Immediate intervention required. High probability of self-harm ideation.
Level 2	**Symptomatic**	depression, lonely	**Moderate Risk:** Targeted distress. Reflects specific internalizing disorders and social isolation.
Level 1	**General Support**	mentalhealth, anxiety	**Baseline/Early Warning:** Broad distress or high-functioning anxiety. Often the "entry point" into the system.

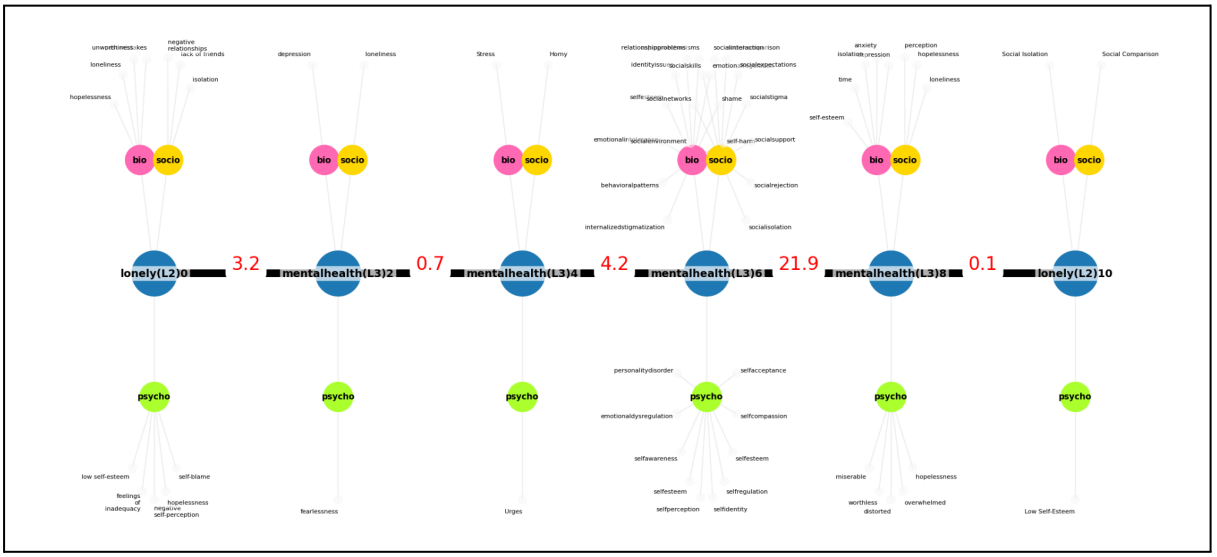


5. The user journey is developed in PostgreSQL for performance and persistence advantage. A sample of 100 users is selected randomly given the resource constraints of local development.

<u>Factor extraction using LLM:</u>

6. The biopsychosocial model is a holistic framework that views health and illness as the result of a dynamic interaction between biological, psychological, and social factors. The model was developed by psychiatrist George Engel in 1977. MedGemma 4b instruct 4bit version is used to extract factors associated with

biopsychosocial models. A temporal knowledge graph is developed connecting the transition graph and the extracted factors. The weights on the edges are the number of days, a temporal feature calculated from the difference of time between the posts and subreddit. Refer to the below image.



Exploratory Data Analysis

From the EDA it could be concluded that

1. There are higher number of psychological factors extracted from the subreddits
2. The distribution of clinical factors and author subreddits are right skewed, indicating either manic episodes, urgency or high social engagement due to positive feedback
3. The derived archetypes from clustering represent predictability with low complexity patterns, noisy medium complexity and sparse high complexity.
4. Velocity (the speed to move from one state to another) has strong correlation with the archetype
5. The factor dimensions has sparse correlation with velocity indicating dense local patterns as to global
6. 60% transitions are stagnant (meaning no movement between states) in comparison to the average transitions

Modelling:

Baseline Modelling

dst_state src_state	anxiety(L3)	depression(L2)	lonely(L2)	mentalhealth(L3)	suicidewatch(L1)
anxiety(L3)	0.662921	0.168539	0.033708	0.101124	0.033708
depression(L2)	0.058511	0.643617	0.063830	0.047872	0.186170
lonely(L2)	0.018182	0.200000	0.654545	0.072727	0.054545
mentalhealth(L3)	0.232143	0.089286	0.071429	0.553571	0.053571
suicidewatch(L1)	0.013575	0.140271	0.013575	0.009050	0.823529

From the baseline transition matrix, it could be concluded that

1. There are sink states, for example L3 -> L3 where the authors seem to be stuck and has the highest probability.
2. The purpose of GNN, a better probabilistic model is to get more better predictions given the additional features of Biopsychosocial factors

Given the transition graph, weights of edges and the factors a predictive modelling could be conducted to predict the path or next link or next node of the user. This is a form of prognosis modelling to help faster and automated clinical interventions and decision making. GCN or GAT type model architecture could be selected for model development.